

Deterministic Certification for the Bounded Normal Mean

An Adaptive-Support Computation with an Independent Verifier

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Research release v0.1.0 – August 2026

Abstract

We study estimation of $\theta \in [-m, m]$ from $Y \sim \mathcal{N}(\theta, 1)$ under squared error. The outer computation follows Kempthorne’s 1987 exchange algorithm: optimize a finitely supported prior, maximize the risk of its Bayes rule over the continuous parameter interval, and add a maximizing support point. Our contribution is narrower. We give a deterministic, grid-free-in-parameter implementation, derive a posterior-flow enclosure for the curvature of the risk, and provide a materially independent verifier based on Arb ball arithmetic, direct Gaussian-mixture ratios, interval quadrature remainders, Gaussian tail bounds, and branch-and-bound. For $m = 4$, the verifier proves that the minimax risk lies in $[0.8150887565050, 0.8150887615302]$, a gap below 5.026×10^{-9} . The common-mean curvature enclosure is independently derived here, but the literature audit has not established priority and its measured reduction in branch nodes is small. We therefore make no claim to a new general least-favorable-prior algorithm or a new generic interval-arithmetic principle. The main result is an independently executable, deterministic certificate for the bounded-normal computation.

Keywords: bounded normal mean; minimax estimation; least favorable prior; interval arithmetic; branch-and-bound; computer-assisted proof.

JEL codes: C14, C44, C63. **MSC 2020:** 62C20, 65G20, 90C26.

1 Introduction

The bounded-normal-mean problem is a small statistical model with an unusually stringent computational lesson. A numerical estimator is easy to produce, but a machine-checkable statement about its worst-case risk requires a global calculation over a continuum. Let

$$Y = \theta + Z, \quad Z \sim \mathcal{N}(0, 1), \quad \theta \in [-m, m],$$

and use loss $(d(Y) - \theta)^2$. The objective is the minimax value

$$v^* = \inf_d \sup_{\theta \in [-m, m]} \mathbb{E}_\theta \{d(Y) - \theta\}^2.$$

Montiel Olea and Zubova [13] recently gave a stochastic mirror-ascent method with an ex ante equally spaced parameter grid and finite-sample probability guarantees. Their paper prompted the present project. Our question is different: after a candidate finitely supported prior has been found, can a deterministic program certify its Bayes risk and the global risk of its Bayes rule without a parameter grid or Monte Carlo?

The answer is yes for the scalar bounded-normal problem. The final $m = 4$ artifact contains three nonnegative support levels, outward-rounded quadrature and tail enclosures, and a continuous branch-and-bound proof. A second program reconstructs the calculation from the compact certificate without importing the discovery solver.

Roadmap. Section 2 separates inherited theory, independent derivations, and implementation. Section 3 proves the exact directional derivative and minimax bracket. Section 4 develops the posterior-flow representation. Sections 5 and 6 give the continuous separator and adaptive procedure. Section 7 describes the independent verifier. Sections 8–9 report tests and compare with Montiel Olea and Zubova. The appendices supply all proof algebra, numerical error bounds, lineage details, and exact reproduction commands.

Contribution stated narrowly. This paper does not introduce adaptive atom insertion, joint optimization of weights and locations, pruning, the directional derivative, or the Bayes-risk/maximum-risk stopping bracket. Those ideas are already present in Kempthorne [10]. Nor does it claim novelty for interval arithmetic, affine arithmetic, semiconvex branch-and-bound, or dependency preservation. The surviving contribution is a specialized deterministic implementation and an independent certificate. The common-mean enclosure in Lemma 1 was derived in this project and was not found in the audited sources, but it is an elementary consequence of posterior moment identities and gave only a modest node reduction. We make no priority claim for it.

2 Exact contribution and relation to prior work

This section states lineage before technical results become operative. Table 1 records the status used throughout the paper.

Table 1: Claim lineage. “Derived here” does not assert literature priority.

Object	Classification	Closest source and exact relation
Adaptive support insertion; reoptimization of masses and locations; zero-mass or coincident-point reduction	Prior art	Kempthorne [10, pp. 175–177], Steps 1–4 and discussion.
Directional derivative and equalizer/stopping condition	Prior art	Kempthorne [10, Theorems 2.1–2.2, pp. 172–174]; the exact Gaussian algebra is supplied here for auditability.
Finite-support structure	Prior art	Kempthorne [10, Theorem 2.4, p. 175], with later structural work including [4, 5].
Deterministic global optimization of fixed finite prior parameterizations	Prior art	Gourdin, Jaumard, and MacGibbon [8, Sections 2–3].
Stochastic grid-based approximation and Wasserstein/estimator convergence	Montiel Olea–Zubova	Theorems 1–4 of [13]; not reproduced or claimed here.
Posterior-flow identities $\mu' = V$, $V' = \kappa_3$	Standard exponential-family algebra	Re-derived in Proposition 2.
Sharp support bound for κ_3	Prior art	Equivalent to Sharma, Gupta, and Kapoor [15, Theorem 2.2 and (2.20)–(2.21)].
Common-mean product enclosure	Derived here; no priority claim	Lemma 1; no exact antecedent located, no priority claim, and only a four-node reduction at $m = 4$.
Arb verifier and $m = 4$ certificate	Implementation result	Separate direct-ratio formula, Simpson remainder, tail bound, and branch-and-bound; source and negative tests accompany the paper.

Casella and Strawderman [2] establish foundational exact results for small bounds. Feldman and Brown [7] and Donoho, Liu, and MacGibbon [3] develop numerical and risk-bound calculations. Gourdin et al. [8] use theoretically convergent Lipschitz and decomposition global optimization for finite-dimensional parameterizations. Kent et al. [11] give a modified Frank–Wolfe method in probability space with oracle assumptions, while Guggenberger and Huang [9] use fictitious play and grid-based best responses in a different minimax-regret problem. Aradillas Fernández et al. [1] prove epsilon-minimax guarantees for mixtures over finitely many decision rules and assume a worst-case-risk oracle. None of these sources, as audited, supplies the present outward-rounded continuous bounded-normal separator and independent certificate as an end-to-end artifact. This is a statement about located content, not a claim of exhaustive priority.

Interval arithmetic itself is classical [14]. Everitt [6] explains dependency over-enclosure and affine arithmetic in statistical computation, and Maruyama and Matsuda [12] use interval arithmetic in a computer-assisted minimaxity proof for a fixed half-Cauchy Bayes estimator. Those precedents preclude a generic “dependency preservation in statistics” claim.

3 The bounded-normal minimax problem

This section defines every statistical object and proves the certificate from the original mixture formulas.

Definition 1 (Symmetric finite prior). *Let $\pi = \sum_{j=1}^J w_j \delta_{\theta_j}$, where $\theta_j \in [-m, m]$, $w_j > 0$, and $\sum_j w_j = 1$. Symmetry means that $\pi(A) = \pi(-A)$ for every Borel set A . Define, for $r = 0, 1, 2$,*

$$f_r(y) = \sum_{j=1}^J w_j \theta_j^r \phi(y - \theta_j), \quad \phi(x) = (2\pi)^{-1/2} e^{-x^2/2}.$$

Since $f_0(y) > 0$, the posterior mean and Bayes estimator are

$$\delta_\pi(y) = \frac{f_1(y)}{f_0(y)}.$$

The frequentist risk at $t \in [-m, m]$ and the Bayes risk are

$$r(t, \delta_\pi) = \int_{\mathbb{R}} \{\delta_\pi(y) - t\}^2 \phi(y - t) dy, \quad (1)$$

$$b(\pi) = \int_{\mathbb{R}} \left\{ f_2(y) - \frac{f_1(y)^2}{f_0(y)} \right\} dy. \quad (2)$$

Fubini's theorem gives the identity

$$b(\pi) = \sum_{j=1}^J w_j r(\theta_j, \delta_\pi). \quad (3)$$

The integrands are dominated because $|\delta_\pi(y)| \leq m$.

Proposition 1 (Exact feasible directional derivative). *For $t \in [-m, m]$ and $\pi_\alpha = (1 - \alpha)\pi + \alpha\delta_t$,*

$$\left. \frac{d}{d\alpha} b(\pi_\alpha) \right|_{\alpha=0+} = r(t, \delta_\pi) - b(\pi).$$

Proof. Write $h_r(y) = t^r \phi(y - t) - f_r(y)$, so that $f_{r,\alpha} = f_r + \alpha h_r$. Pointwise differentiation of $f_{2,\alpha} - f_{1,\alpha}^2/f_{0,\alpha}$ gives

$$h_2 - \frac{2f_1 h_1}{f_0} + \frac{f_1^2 h_0}{f_0^2}.$$

Substitute h_r and $\delta_\pi = f_1/f_0$. The terms involving the original mixture equal

$$-f_2 + 2\delta_\pi f_1 - \delta_\pi^2 f_0 = -f_2 + \frac{f_1^2}{f_0},$$

whereas the atom terms equal

$$\{t^2 - 2t\delta_\pi(y) + \delta_\pi(y)^2\} \phi(y - t).$$

Integration yields $r(t, \delta_\pi) - b(\pi)$. Differentiation under the integral is justified in Appendix A; the key facts are bounded support and strict positivity of the Gaussian mixture. $\square$

Corollary 1 (Exact minimax certificate). *For every prior π supported on $[-m, m]$,*

$$b(\pi) \leq v^* \leq U(\pi) := \sup_{t \in [-m, m]} r(t, \delta_\pi),$$

and therefore

$$0 \leq v^* - b(\pi) \leq U(\pi) - b(\pi) = \sup_{t \in [-m, m]} \{r(t, \delta_\pi) - b(\pi)\}.$$

Proof. For any decision rule d , its maximum risk is at least its π -average risk, which is at least the Bayes risk $b(\pi)$. Taking the infimum over d proves the lower bound. The Bayes rule δ_π is an admissible competitor in the minimax infimum, which proves the upper bound. Subtraction and Proposition 1 give the last display. $\square$

Corollary 1, essentially Kempthorne's stopping condition, is the logical core of the computation: a lower enclosure for $b(\pi)$ and an upper enclosure for $U(\pi)$ produce a rigorous interval for v^* .

4 Posterior-flow representation

The separator must control a ratio of Gaussian mixtures without replacing its shared numerator and denominator by independent intervals. This section identifies a small posterior state that retains the relevant dependence.

For an observation y , let

$$p_j(y) = \frac{w_j \exp(\theta_j y - \theta_j^2/2)}{\sum_k w_k \exp(\theta_k y - \theta_k^2/2)}.$$

The omitted common factor $\phi(y)$ cancels. Define the posterior mean, variance, and cumulants

$$\mu(y) = \sum_j p_j(y) \theta_j, \quad V(y) = \sum_j p_j(y) \{\theta_j - \mu(y)\}^2, \quad \kappa_s(y) = \text{sth posterior cumulant}.$$

Proposition 2 (Posterior flow and risk curvature). *For a finite bounded prior, all functions above are analytic and*

$$\mu' = V, \quad V' = \kappa_3, \quad \kappa'_s = \kappa_{s+1}.$$

With $q(t, z) = \mu(t + z) - t$ and $Z \sim \mathcal{N}(0, 1)$,

$$r(t, \delta_\pi) = \mathbb{E} q(t, Z)^2, \tag{4}$$

$$r'(t, \delta_\pi) = 2\mathbb{E}\{q(t, Z)(V(t + Z) - 1)\}, \tag{5}$$

$$r''(t, \delta_\pi) = 2\mathbb{E}\{(V(t + Z) - 1)^2 + q(t, Z)\kappa_3(t + Z)\}. \tag{6}$$

Proof. The probabilities form a finite natural exponential family with natural parameter y . Successive derivatives of

$$\log \left(\sum_j w_j e^{\theta_j y - \theta_j^2/2} \right)$$

produce the cumulants. Further, $\partial_t q = V(t + z) - 1$ and $\partial_t V = \kappa_3$. Two differentiations of $\mathbb{E} q^2$ give the displayed formulas. Bounded support supplies an integrable uniform bound for each derivative. $\square$

The only sign-indefinite product in (6) is $q\kappa_3$. The following representation couples its factors.

Lemma 1 (Common-mean representation and enclosure). *Let the posterior be supported on $[\alpha, \beta]$. If $V > 0$, define*

$$\nu = \frac{\mathbb{E}\{\Theta(\Theta - \mu)^2 \mid y\}}{V}.$$

Then $\nu \in [\alpha, \beta]$ and

$$\kappa_3 = V(\nu - \mu).$$

If $t \in [a, b]$, $\mu \in M = [\mu_L, \mu_U]$, and $0 \leq V \leq V_U$, then

$$q\kappa_3 \leq V_U \max_{\mu \in M} \{(\mu - a)_+(\beta - \mu), (b - \mu)_+(\mu - \alpha)\}, \tag{7}$$

$$q\kappa_3 \geq -V_U \max_{\mu \in M} \{(\mu - a)_+(\mu - \alpha), (b - \mu)_+(\beta - \mu)\}. \tag{8}$$

Each maximum is either an endpoint value or the value at the vertex of one quadratic.

Proof. The weights proportional to $p_j(\theta_j - \mu)^2$ are nonnegative and sum to V , hence ν is their mean and lies in the convex hull $[\alpha, \beta]$. Expanding $\mathbb{E}\{\Theta(\Theta - \mu)^2\} = \kappa_3 + \mu V$ proves the identity. For a positive product, if $\mu \geq t$ choose the largest feasible $\nu = \beta$; if $\mu \leq t$ choose $\nu = \alpha$. Maximizing over $t \in [a, b]$ gives (7). The opposite choices of ν give the most negative product and (8). The case $V = 0$ has $\kappa_3 = 0$ and follows by continuity. $\square$

Equality and degeneracy. Equality in the identity holds by construction. An envelope endpoint can be attained only when the posterior squared-deviation weights put their mean ν at the relevant support endpoint and when $V = V_U$; $V = 0$ collapses all terms to zero. These conditions are useful diagnostics but are not assumed by the solver.

What the bound does not retain. The relaxation still separates V from μ through V_U and therefore is not the exact joint range. It is stronger than a Cartesian product of independent intervals for q and κ_3 , but it is not advertised as the smallest possible state. The ablation in Section 8 measures the remaining loss.

5 Certified continuous separation

This section turns point-risk and curvature enclosures into a global upper bound. The final verifier uses direct mixture-ratio interval boxes rather than the specialized common-mean formula, providing formula-level independence.

Lemma 2 (Semiconvex secant majorant). *Let f be twice differentiable on $[a, b]$ and $f'' \geq -K$ there, with $K \geq 0$. For $x = a + s(b - a)$, $s \in [0, 1]$,*

$$f(x) \leq (1 - s)f(a) + sf(b) + \frac{K}{2}(b - a)^2 s(1 - s).$$

Consequently, the cell maximum is bounded by the maximum of the two endpoints and, when its vertex lies in $[0, 1]$, the vertex value of the quadratic on the right.

Proof. The function $f(x) + Kx^2/2$ is convex. Apply the defining secant inequality for a convex function and collect the quadratic terms. No assumption about the sign of f' is required. $\square$

Proposition 3 (Risk quadrature enclosure). *Fix $t \in [-m, m]$ and $A > 0$. Composite Simpson quadrature on $[-A, A]$, with an interval enclosure of the fourth derivative on every panel, encloses the truncated risk integral. The omitted tails satisfy*

$$0 \leq \int_{|z| > A} q(t, z)^2 \phi(z) dz \leq 8m^2 \frac{\phi(A)}{A}.$$

Proof. The panel error is $-h^5 H^{(4)}(\xi)/2880$ when h denotes the full panel width. Replacing $H^{(4)}(\xi)$ by its interval range is inclusion monotone. Since $|\mu - t| \leq 2m$, the tail is at most $4m^2 \Pr(|Z| > A)$; Mills' inequality $\Pr(|Z| > A) \leq 2\phi(A)/A$ gives the result. $\square$

The verifier obtains a lower enclosure for (6) by partitioning $[-A, A]$ in z , evaluating the common posterior ratio on the box $t \in [a, b]$, and adding a two-sided curvature-tail enclosure. If $0 \leq V \leq m^2$, then

$$|(V - 1)^2 + q\kappa_3| \leq \max\{1, (m^2 - 1)^2\} + 4m^4,$$

because $|q| \leq 2m$ and $|\kappa_3| \leq 2mV \leq 2m^3$. This gives an explicit tail bound for (6).

Theorem 1 (Verifier implication). *Suppose an outward-rounded computation returns bounds satisfying*

$$B_L \leq b(\pi) \quad \text{and} \quad U_U \geq \sup_{t \in [-m, m]} r(t, \delta_\pi).$$

Then

$$v^* \in [B_L, U_U], \quad 0 \leq v^* - B_L \leq U_U - B_L.$$

If $U_U - B_L \leq \varepsilon$, the Bayes rule is certifiably ε -minimax in value.

Proof. This is Corollary 1 with interval endpoints weakened outward. The theorem is a certification corollary, not a new minimax principle. $\square$

6 Adaptive computational procedure

The outer loop below is Kempthorne's algorithm specialized to symmetric bounded-normal priors. Symmetry reduces the search to nonnegative levels; a positive level represents an equal pair at $\pm t$, while level zero represents one atom.

Fast mode. The discovery solver uses float64, deterministic Gauss-Hermite quadrature, warm starts, quasi-Newton-style mass and location updates, aggressive pruning, and the common-mean curvature enclosure. Its bounds are diagnostics, not certificates. No parameter grid is fixed in advance; branch cells are created adaptively.

Adaptive support computation followed by certification. Given $m > 0$, a requested gap $\varepsilon > 0$, and an initial symmetric prior:

1. reoptimize retained level masses;
2. optionally optimize interior support locations while keeping them ordered;
3. prune zero masses and merge numerically coincident levels;
4. compute a continuous global risk maximizer or upper bound on $[0, m]$;
5. if the numerical separation gap is not small, add the symmetric pair at the maximizing level and return to step 1;
6. serialize only m , levels, masses, verifier settings, and claims;
7. run the independent outward-rounded verifier; and
8. return the prior, Bayes rule, $[B_L, U_U]$, and report only if the verifier proves $U_U - B_L \leq \varepsilon$. Otherwise refine and repeat.

Figure 1: The outer loop is inherited from Kempthorne; certification is a separate post-processing stage.

Certificate mode. The final computation uses 192-bit Arb balls. The verifier recomputes posterior ratios, Simpson sums and remainder intervals, tail bounds, cell curvatures, and continuous branch coverage. It accepts no discovery-solver branch tree.

Convergence statement. Kempthorne [10, Theorem 3.1] proves asymptotic convergence under assumptions A1–A5 for the abstract exchange procedure. The present implementation does not prove a finite iteration bound for finding a certifiable prior. Its guarantee is conditional and post hoc: whenever the independent verifier accepts, Theorem 1 proves the reported gap. This distinction prevents a numerical success from being misstated as an unconditional finite-termination theorem.

7 Independent certificate and verifier

The compact certificate contains m , nonnegative support levels, total level masses, arithmetic precision, quadrature and branch settings, and claimed risk intervals. It contains no estimator samples, parameter grid, Monte Carlo draws, or trusted branch partition.

The verifier performs the following checks.

1. It rejects nonfinite input, support outside $[0, m]$, unordered or duplicate levels, nonpositive masses, and masses that do not sum exactly to one as decimal inputs.
2. It reconstructs both atoms at every positive level and computes the posterior directly from $w_j \exp(\theta_j y - \theta_j^2/2)$.
3. It evaluates the risks at all support levels with independent composite Simpson code, fourth-derivative intervals, and Gaussian tails, then averages them to enclose the Bayes risk.
4. It starts from the complete interval $[0, m]$. Each branch cell receives a curvature lower bound and the exact semiconvex quadratic majorant. Only a cell whose upper bound is within the work tolerance of the incumbent may be pruned.
5. It checks that the supplied claim intervals contain the recomputation and that the final gap is below the requested tolerance.

The verifier imports neither `solver_fast.py` nor `solver_certified.py`. It does share the mathematical specification: Gaussian density, posterior mean, risk, and the semiconvex theorem. Complete formula independence would be impossible without changing the quantity being verified. The implementation of those formulas and the arithmetic library are independent.

Negative tests. Structural tests reject negative and zero masses, invalid normalization, duplicate and out-of-range support, nonfinite values, inverted claims, false gap arithmetic, and inadequate precision. Numerical recomputation rejects a validly normalized weight perturbation, a support perturbation, and a structurally consistent understatement of worst-case risk. An omitted branch cannot be injected because no serialized branch list is accepted: branch coverage is regenerated from $[0, m]$.

Trusted computing base. The final proof trusts CPython, python-flint 0.8.0, and the Arb/FLINT implementation beneath it, plus the source code of the verifier. The discovery optimizer and NumPy are outside the final trusted base. This is a computer-assisted proof, not a formal proof in a proof assistant.

8 Numerical results and ablations

All timings in this section are single-process measurements on the local macOS 15.7.3 arm64 machine and are descriptive rather than hardware-normalized complexity claims.

8.1 The certified result for $m=4$

The final prior has nonnegative levels

$$(0.5984699814893856, 2.1264117002421385, 4)$$

and total level masses

$$(0.36699366016033996, 0.3994285510645993, 0.23357778877506074).$$

Each positive level is split equally between its positive and negative atom. The independent 192-bit verification used tail cutoff $A = 8$, 2048 Simpson panels per point risk, 512 curvature cells, and 567 branch nodes. It took 49.68 seconds in the recorded clean run.

Table 2: Outward-rounded $m = 4$ certificate. Displayed endpoints are rounded outward from the longer Arb output.

Quantity	Independently recomputed enclosure
Bayes risk	$[0.8150887565050, 0.8150887565548]$
Worst-case risk	$[0.8150887565301, 0.8150887615302]$
Minimax risk	$[0.8150887565050, 0.8150887615302]$
Certified gap upper bound	$5.02517E - 9$
Branch nodes	567
Arithmetic	python-flint Arb balls, 192 bits

The verifier’s Bayes-risk width is 4.98×10^{-11} . The branch envelope uses a 5.0×10^{-9} work width. The certified minimax-gap upper bound is 5.02517×10^{-9} , below 10^{-8} .

8.2 Fast deterministic sweep

Table 3 reports float64 discovery results. They are not certificates except where a separate certificate is stated.

8.3 Dependency ablation

Table 4 holds the prior and endpoint-risk evaluator fixed. The two generic global bounds exhaust the 5001-node budget. Natural interval, one-noise affine, Cartesian posterior-flow, and common-mean flow bounds terminate. These rows are float64 diagnostics and do not replace the Arb certificate.

The common-mean formula reduces the Cartesian-flow tree from 281 to 277 nodes. Its much shorter runtime in this implementation comes primarily from vectorized engineering, not a large mathematical tightening. Across cell widths from approximately 3.87 to 10^{-5} , the ratio of common-mean to Cartesian curvature width ranged from 0.864 to 0.996 . This evidence does not support a claim that the enclosure is a broadly superior new bound.

Table 3: Adaptive-support float64 sweep; no ex ante parameter grid.

m	levels	atoms	Bayes risk	float gap	seconds
0.25	1	2	0.0588158765	0.00e+00	0.003
0.5	1	2	0.1989864336	0.00e+00	0.000
1	1	2	0.4495995092	0.00e+00	0.001
1.5	2	3	0.5561720123	3.78e-10	0.020
2	2	3	0.6448008729	3.47e-09	0.030
3	3	5	0.7506254421	3.62e-09	0.076
4	3	6	0.8150887565	2.64e-09	0.187
5	4	7	0.8573009664	3.71e-09	0.218
8	6	11	0.9231345345	6.26e-09	5.149

Table 4: $m = 4$, target separator width 10^{-8} .

Separator	nodes	exhausted	width	seconds
global Lipschitz	5001	yes	2.34e-01	0.115
global semiconvex	5001	yes	5.09e-04	0.113
natural interval	291	no	8.58e-09	2.248
one-noise affine	427	no	9.95e-09	2.208
Cartesian flow	281	no	5.78e-09	3.170
common-mean flow	277	no	4.57e-09	0.076

8.4 Hostile tests

With recorded seed 20260729, 100 randomized valid symmetric priors and parameter intervals produced no dense-sample violation of the float64 curvature enclosure. Separate tests covered a 10^{-8} pair of nearly coincident levels, a level mass of 10^{-12} , boundary and zero atoms, $y = \pm 50, \pm 100$ posterior ratios, narrow cells, and Gaussian tail cutoffs 6, 7, 8, 9. A deliberate 10^{-6} symmetry perturbation changed positive and negative risks and therefore was not silently accepted as symmetric. These tests are falsification evidence, not proof; the proof comes from outward-rounded inclusion.

9 Comparison with Montiel Olea and Zubova

The comparison must account for different objectives. Montiel Olea and Zubova [13] use stochastic mirror ascent on an equally spaced grid. Their theory controls discretization and stochastic optimization error, proves Wasserstein convergence of approximate priors, and proves convergence of the resulting estimators. The method is expressed through broadly applicable first-order stochastic optimization. The present solver instead exploits scalar Gaussian posterior identities and is designed for deterministic certification of this one model.

The supplied source archive did not contain the stochastic implementation, approximate-prior arrays, random seed, or hardware specification. We therefore could not reproduce their random path or claim a hardware-matched speed ratio. We reconstructed their stated tuning formulas and digitized their source risk figure using a checksum-locked pixel calibration. The deterministic values for $m = 1, 2, 3, 4$ lie within the displayed source-figure intervals to the declared pixel error.

At $m = 4$ the present fast solver took 0.25 seconds in the clean reproduction, and the independent certificate took 49.68 seconds. The target paper does not report an $m = 4$ runtime under specified hardware, so a “100 times faster” statement is not identified and is not made.

The present computation can independently validate the bounded-normal minimax value used as an ingredient in their LP/VAR application. It does not reproduce the full empirical pipeline, and it does not show that stochastic mirror ascent is mistaken or inferior in general. It may serve as a specialized alternative, benchmark, or supplementary certificate.

Table 5: Scope comparison.

Feature	Montiel Olea–Zubova	Present project
Parameter representation	Fixed equally spaced grid	Adaptive continuous branch cells; finite active support
Optimization	Stochastic mirror ascent over grid weights	Kemphorne exchange; float64 mass/location reoptimization
Integration	Monte Carlo supergradient and reported risk evaluation	Deterministic Gauss–Hermite discovery; interval Simpson verification
Guarantee	High-probability approximation and convergence theory	Post-hoc deterministic value certificate for each accepted prior
Continuous separation	Lipschitz grid correction	Outward-rounded branch-and-bound on $[0, m]$
Generality	First-order method with broader optimization relevance	Specialized scalar bounded-normal implementation
Reproducibility from supplied archive	Manuscript and raster figures; no code, seed, arrays, or hardware	Source, lockfile, certificate, and independent verifier

10 Limitations, scope, and conclusion

The result is deliberately narrow. The final certificate concerns the one-dimensional normal location model with unit variance, squared loss, bounded symmetric parameter space, and a supplied finite symmetric prior. The verifier does not prove that the discovery solver always finds such a prior in finite time. Complexity bounds for the specialized branch tree are not established. The certificate relies on a software trusted base rather than a proof assistant. Only $m = 4$ receives the final Arb certificate in this version; the remaining sweep is float64 evidence.

The prior-art audit did not locate an end-to-end deterministic, outward-rounded continuous certificate for this bounded-normal computation. It did locate nearly all of the outer algorithmic architecture in Kemphorne, fixed-dimensional deterministic global optimization in Gourdin et al., generic dependency-aware arithmetic in the interval literature, and computer-assisted minimaxity in Maruyama–Matsuda. The common-mean enclosure is valid but empirically only modestly tighter in node count. The paper therefore treats the independent executable certificate—not a broad new algorithmic theorem—as its principal deliverable.

Declarations

Code and reproducibility. The public repository contains the compact $m = 4$ certificate, the discovery and construction implementations, the independent Arb verifier, the test suite, and locked dependency metadata. Appendix D gives the release verification commands.

Software and hardware. The clean environment used CPython 3.13.5, NumPy 2.3.2, Pillow 11.3.0, and python-flint 0.8.0 on macOS 15.7.3 arm64. The exact processor marketing name and memory capacity were not recorded, so no hardware-normalized timing claim is made.

Conflict of interest. The author declares no conflict of interest.

Funding. No project-specific external funding was identified in the available author metadata.

Acknowledgements. The project was motivated by the bounded-normal computation of Montiel Olea and Zubova. This acknowledgement does not imply their endorsement of the present implementation or certificate.

AI-assisted research disclosure. Large language models assisted with literature search, theorem exploration, proof drafting, code generation, adversarial testing, and exposition. They are not authors. Responsibility for every claim remains with the human author. Reported computational results are accompanied by executable verification and remain subject to independent human review.

A Proof details

A.1 Differentiation of Bayes risk

Fix $\alpha_0 < 1$. For $\alpha \in [0, \alpha_0]$, $f_{0,\alpha} \geq (1 - \alpha_0)f_0$. Directly expanding the differentiated integrand in Proposition 1 reduces it to the sum of $-f_2 + f_1^2/f_0$ and $(t - \delta_\pi)^2\phi(\cdot - t)$. The first absolute value is bounded by $f_2 + m^2f_0$, and the second by $4m^2\phi(\cdot - t)$. Both are integrable. Dominated convergence therefore justifies the right derivative. The same argument applies to a direction given by any bounded-support probability measure after integrating the atom formula in t .

A.2 Fourth derivative used by Simpson's rule

Let $u(z) = q(t, z)^2$. With derivatives taken in z ,

$$\begin{aligned} u_0 &= q^2, \\ u_1 &= 2qV, \\ u_2 &= 2(V^2 + q\kappa_3), \\ u_3 &= 2(3V\kappa_3 + q\kappa_4), \\ u_4 &= 2(3\kappa_3^2 + 4V\kappa_4 + q\kappa_5). \end{aligned}$$

Because $\phi' = -z\phi$, repeated product differentiation gives

$$(u\phi)^{(4)} = \phi[u_4 - 4zu_3 + 6(z^2 - 1)u_2 + 4(3z - z^3)u_1 + (z^4 - 6z^2 + 3)u_0].$$

Every posterior quantity is evaluated on the same (t, z) box.

A.3 Curvature tail

For posterior support in $[-m, m]$, $0 \leq V \leq m^2$ by the Bhatia–Davis inequality. Also

$$|\kappa_3| = |\mathbb{E}(\Theta - \mu)^3| \leq 2m \mathbb{E}(\Theta - \mu)^2 = 2mV.$$

Thus the absolute curvature state is no greater than

$$C_m = \max\{1, (m^2 - 1)^2\} + 4m^4.$$

Outside $[-A, A]$, the magnitude omitted from (6) is at most $2C_m \Pr(|Z| > A) \leq 4C_m\phi(A)/A$.

A.4 Branch pruning invariant

At every time the active heap plus already pruned cells covers $[0, m]$. An active cell has a proved upper bound. A cell is pruned only if its bound is no larger than the current incumbent plus the work tolerance. Therefore the global supremum is bounded by the maximum of the active-cell bounds and the incumbent plus the work tolerance. This maximum is exactly the value reported by the verifier. The use of floating-point heap priorities affects only processing order; each retained mathematical bound is an Arb ball.

B Implementation and verification details

B.1 Error budget at $m=4$

The independently recomputed Bayes-risk enclosure width is 4.9777×10^{-11} . Of this, the reported Simpson-remainder width is 4.9696×10^{-11} , the Gaussian tail upper bound is 8.084×10^{-14} , and arithmetic rounding in the Simpson sum is below 3.1×10^{-40} . At risk points encountered by separation, the largest fourth-order remainder width is 6.5690×10^{-11} and the largest tail upper bound is 8.084×10^{-14} . The branch envelope width is 5.0×10^{-9} .

B.2 Independent reconstruction

The discovery solver computes posterior states with vectorized NumPy and a specialized flow tube. The final verifier instead:

1. constructs Arb balls from decimal strings;
2. evaluates each mixture exponential directly;
3. forms posterior central moments through order five;
4. implements a separate Simpson panel loop;
5. uses direct interval boxes for curvature rather than the common-mean flow formula; and
6. rebuilds its own branch tree from $[0, m]$.

This separation is material but not absolute: both programs implement the same mathematical definition of the risk.

B.3 Failure modes examined

Independent intervals for f_0 , f_1^2 , and $1/f_0$ repeatedly count the same uncertainty and become particularly wide in the tails. A global Lipschitz bound scales only linearly with cell width and exhausted the 5001-node budget even at tolerance 10^{-4} . A global curvature bound also exhausted that budget. One-noise affine arithmetic helps on narrow cells but was not uniformly tighter than the posterior-flow representation. These are empirical no-go results for the tested representations, not impossibility theorems for all refinements of those methods.

C Detailed prior-art comparison

Kemphorne’s Section 3 begins with a finite prior, locally maximizes Bayes risk over weights and locations, globally maximizes the Bayes-rule risk, stops when maximum risk equals Bayes risk, and otherwise inserts the maximizer [10, pp. 175–177]. He explicitly notes that zero weights or coincident locations can reduce support. His implementation brackets local risk maxima by a parameter grid and refines them by golden-section search [10, p. 177]; it is not an outward-rounded continuous certificate.

Gourdin et al. terminate their Lipschitz global optimizer when the gap between an upper bound and incumbent is below a chosen tolerance [8, pp. 19–21]. Their bounded-normal discussion fixes a finite-dimensional prior form justified for a small range of m [8, pp. 25–26]. This is important deterministic global optimization prior art, but it is not the same as adaptive support generation with an interval-certified continuous risk oracle for general m .

Dytso et al. [5] give support-cardinality reductions under a finite observation alphabet and use projected gradient ascent in numerical examples. The scalar Gaussian observation has an infinite alphabet, so their stated finite-alphabet cardinality theorem does not directly supply this solver. Kent et al. [11] assume derivative and local linear minimization oracles and use stochastic subroutines; they do not instantiate the bounded-normal separator. Guggenberger and Huang [9] obtain iterationwise brackets in a finite/discretized fictitious-play problem but calculate nature’s best response by grid search.

Maruyama and Matsuda [12] are the closest located precedent for interval-assisted normal-mean minimaxity. They verify inequalities for the Bayes estimator under a fixed half-Cauchy prior in a multivariate problem; they do not search for a least-favorable bounded scalar prior or certify the Bayes/maximum-risk gap here.

The audit was source-grounded but cannot prove a universal negative over all literature. Closer antecedents may exist, so no exhaustive priority claim is made.

D Reproducibility protocol

From the public repository root, create the locked environment and run the deterministic test suite with

```
uv sync --frozen
uv run python -m unittest discover -s tests -v
```

The lockfile pins NumPy, Pillow, and python-flint. No global installation or administrator access is required.

For the shortest independent check after environment setup:

```
uv run python src/verification/verify_certificate_arb.py \  
certificates/final/m4_epsilon_1e-8_arb.json
```

The command exits nonzero if the certificate fails. Exact file hashes appear inside the compact certificate where they are part of the verification contract. A separate corruption audit is available as

```
uv run python -m scripts.run_corruption_audit \  
--certificate certificates/final/m4_epsilon_1e-8_arb.json \  
--output /tmp/bounded-normal-corruption-audit.json
```

The public repository is <https://github.com/Maar10Herr/Bounded-Normal-Minimax-Certificate-Public>.

References

- [1] Andrés Aradillas Fernández, José Blanchet, José Luis Montiel Olea, Chen Qiu, Jörg Stoye, and Lezhi Tan. Epsilon-minimax solutions of statistical decision problems, 2026. Version 2, 8 January 2026.
- [2] George Casella and William E. Strawderman. Estimating a bounded normal mean. *The Annals of Statistics*, 9(4):870–878, 1981.
- [3] David L. Donoho, Richard C. Liu, and Brenda MacGibbon. Minimax risk over hyperrectangles, and implications. *The Annals of Statistics*, 18(3):1416–1437, 1990.
- [4] Alex Dytso, Ronit Bustin, H. Vincent Poor, and Shlomo Shamai. On the structure of the least favorable prior distributions. In *2018 IEEE International Symposium on Information Theory*, pages 1081–1085, 2018.
- [5] Alex Dytso, Mario Goldenbaum, H. Vincent Poor, and Shlomo Shamai. A dimensionality reduction method for finding least favorable priors with a focus on bregman divergence. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, volume 151 of *Proceedings of Machine Learning Research*, pages 7355–7372, 2022.
- [6] Richard G. Everitt. Efficient importance sampling in low dimensions using affine arithmetic. *Computational Statistics*, 33(1):1–29, 2018.
- [7] Israel Feldman and Lawrence D. Brown. The minimax risk for estimating a bounded normal mean. Technical report, Cornell University Statistics Center, 1989. Technical report.
- [8] Eric Gourdin, Brigitte Jaumard, and Brenda MacGibbon. Global optimization decomposition methods for bounded parameter minimax risk evaluation. *SIAM Journal on Scientific Computing*, 15(1):16–35, 1994.
- [9] Patrik Guggenberger and Jiaqi Huang. On the numerical approximation of minimax regret rules via fictitious play, 2025.
- [10] Peter J. Kempthorne. Numerical specification of discrete least favorable prior distributions. *SIAM Journal on Scientific and Statistical Computing*, 8(2):171–184, 1987.
- [11] Carson Kent, Jiajin Li, José Blanchet, and Peter W. Glynn. Modified frank wolfe in probability space. In *Advances in Neural Information Processing Systems*, volume 34, pages 14448–14462, 2021.
- [12] Yuzo Maruyama and Takeru Matsuda. Minimaxity under the half-cauchy prior. *Journal of Multivariate Analysis*, 208:105431, 2025.
- [13] José Luis Montiel Olea and Ekaterina Zubova. Approximate minimax estimation of a bounded normal mean via stochastic mirror ascent, 2026. Version 1, 7 July 2026.
- [14] Ramon E. Moore, R. Baker Kearfott, and Michael J. Cloud. *Introduction to Interval Analysis*. SIAM, Philadelphia, 2009.
- [15] Ritesh Sharma, Mukesh Gupta, and G. Kapoor. Some better bounds on the variance with applications. *Journal of Mathematical Inequalities*, 4(3):355–363, 2010.